

Web Application Vulnerability Testing

1. Introduction

This task focuses on identifying common web application vulnerabilities using industry-standard security testing tools. The objective is to understand how vulnerabilities occur, how attackers exploit them, and how they can be mitigated. The assessment was performed in a controlled environment using a deliberately vulnerable web application.

2. Tools Used

- Burp Suite Community Edition
- OWASP Juice Shop
- Web Browser (Chrome/Firefox)

3. Application Tested

Application Name: OWASP Juice Shop

Type: Deliberately Vulnerable Web Application

Environment: Localhost

4. Methodology

- Studied OWASP Top 10 vulnerabilities
- Configured Burp Suite proxy

- Intercepted HTTP requests
- Tested for SQL Injection
- Tested for XSS
- Documented findings

5. Vulnerabilities Identified

1 . SQL Injection

Payload Used: ' OR '1'='1

Impact: Authentication bypass

Mitigation: Prepared statements, input validation

2 . Cross-Site Scripting (XSS)

Payload Used: <script>alert('XSS')</script>

Impact: Session hijacking, cookie theft

Mitigation: Output encoding, CSP

6. How Burp Suite Works

Burp Suite acts as a man-in-the-middle proxy to intercept and analyze traffic.

7. Conclusion

This task provided hands-on experience in identifying web vulnerabilities in a safe environment.

